

$$\mathbb{Q}(\zeta_7)^+, \mathbb{Q}(\zeta_9)^+$$

the sixteen totally real cubic fields F_0 , by discriminant

disc F_0	49	81	229	257	321	469	473	697	761	785	985	993	1129	1229	1345	1509
k	2	2	2	2	2	2	3	3	2	3	3	3	3	3	3	3
$q = 1$	296	488	1080	1048	1336	1880	1808	2808	2896	2896	2896	2896	3536	3872	3536	3872
$q = k + \alpha$	296	488	840	352	728	616	1208	1688	616	2624	3048	3000	1944	1264	1832	2064
$q = k + 1 + \alpha$	192	256	1672	520	1968	3144	2288	4264	2984	3848	7080	5336	5240	6768	4152	6504

$-\chi(v, v)$, least with $\Omega(v) \neq 0$



-  $F = F_0(\sqrt{-q}) \supset \mathbb{Q}(\sqrt{-1})$: every $q = 1$, and $q = 2 + \alpha$ over $\mathbb{Q}(\zeta_7)^+, \mathbb{Q}(\zeta_9)^+$; lattice minimum 32, attained with $\Omega = 0$
-  $F \supset \mathbb{Q}(\sqrt{-3})$: $q = 2 + \alpha$ at disc $F_0 = 321$, $q = 4 + \alpha$ at 1509; lattice minima 288 and 4608, attained with $\Omega = 0$
-  $192 = -\chi(v_*)$, $v_* = v(-1, 0, 2 - \alpha^2, 0)$, over $\mathbb{Q}(\zeta_7)^+$ with $q = 3 + \alpha$: the least of the forty-eight, attained only at $\pm v_*$
-  $q = 1$: 296 and 488; every class with smaller $-\chi$ lies in $\mathbb{Z} \operatorname{Re} e^{\sqrt{-1}\theta} \oplus \mathbb{Z} \operatorname{Im} e^{\sqrt{-1}\theta}$, where $\Omega = 0$